```
Dataset loaded successfully!
Dataset shape: (891, 11)
```

	PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	Fare	Embarked
0	1	0	3	Braund, Mr. Owen Harris	male	22.0	1	0	A/5 21171	7.2500	S
1	2	1	1	Cumings, Mrs. John Bradley (Florence Briggs Th...	female	38.0	1	0	PC 17599	71.2833	C
2	3	1	3	Heikkinen, Miss. Laina	female	26.0	0	0	STON/O2. 3101282	7.9250	S
3	4	1	1	Futrelle, Mrs. Jacques Heath (Lily May Peel)	female	35.0	1	0	113803	53.1000	S
4	5	0	3	Allen, Mr. William Henry	male	35.0	0	0	373450	8.0500	S

```
Dataset dimensions: (891, 11)

Column names:
['PassengerId', 'Survived', 'Pclass', 'Name', 'Sex', 'Age', 'SibSp', 'Parch', 'Ticket', 'Fare', 'Embarked']

Data types and non-null values:
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 891 entries, 0 to 890
Data columns (total 11 columns):
 #   Column       Non-Null Count  Dtype
---  ------       --------------  -----
 0   PassengerId  891 non-null    int64
 1   Survived     891 non-null    int64
 2   Pclass       891 non-null    int64
 3   Name         891 non-null    object
 4   Sex          891 non-null    object
 5   Age          891 non-null    float64
 6   SibSp        891 non-null    int64
 7   Parch        891 non-null    int64
 8   Ticket       891 non-null    object
 9   Fare         891 non-null    float64
 10  Embarked     891 non-null    object
dtypes: float64(2), int64(5), object(4)
memory usage: 76.7+ KB

Missing values:
PassengerId    0
Survived       0
Pclass         0
Name           0
Sex            0
Age            0
SibSp          0
Parch          0
Ticket         0
Fare           0
Embarked       0
dtype: int64

Duplicate rows: 0

Descriptive statistics:
```

	PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	Fare	Embarked
count	891.000000	891.000000	891.000000	891	891	891.000000	891.000000	891.000000	891	891.000000	891
unique	NaN	NaN	NaN	891	2	NaN	NaN	NaN	681	NaN	3
top	NaN	NaN	NaN	Dooley, Mr. Patrick	male	NaN	NaN	NaN	347082	NaN	S

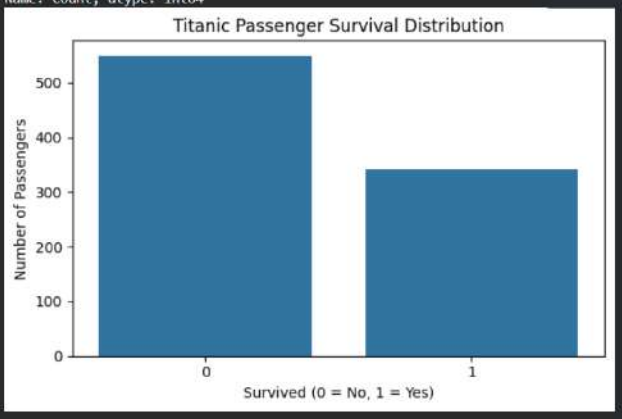
```
Prepared dataset shape: (891, 11)

Missing values after duplicate and target handling:
PassengerId    0
Survived       0
Pclass         0
Name           0
Sex            0
Age            0
SibSp          0
Parch          0
Ticket         0
Fare           0
Embarked       0
dtype: int64
```

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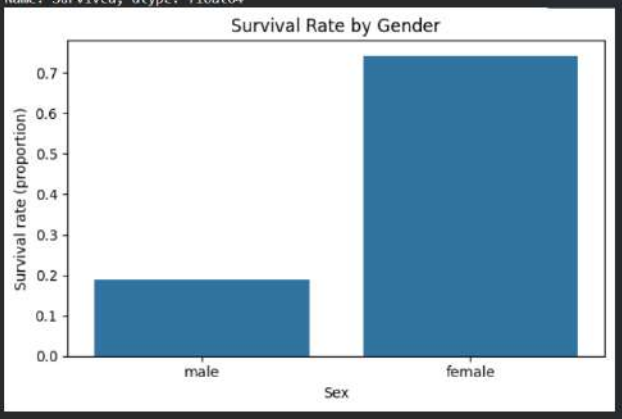
```
Survival counts:
Survived
0    549
1    342
Name: count, dtype: int64
```

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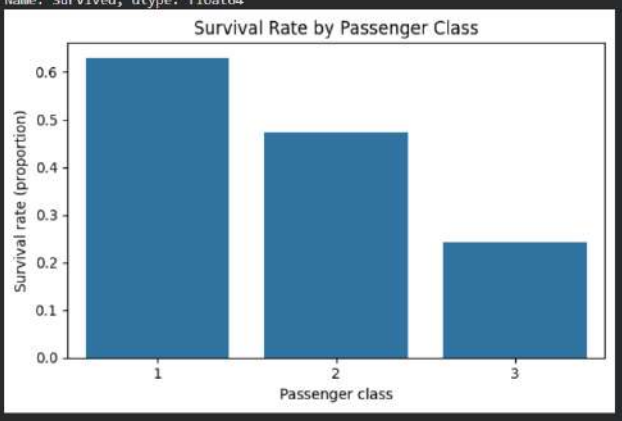
```
Survival rate by gender (%):
Sex
female    74.20
male      18.89
Name: Survived, dtype: float64
```

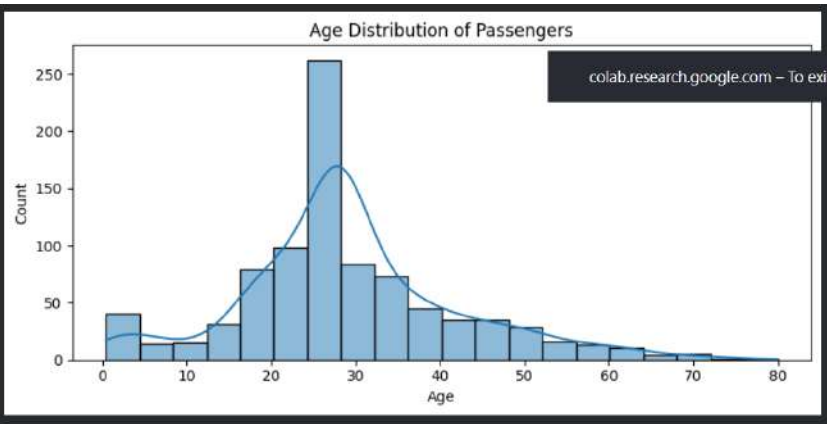
colab.research.google.com – To exit full screen, press Esc



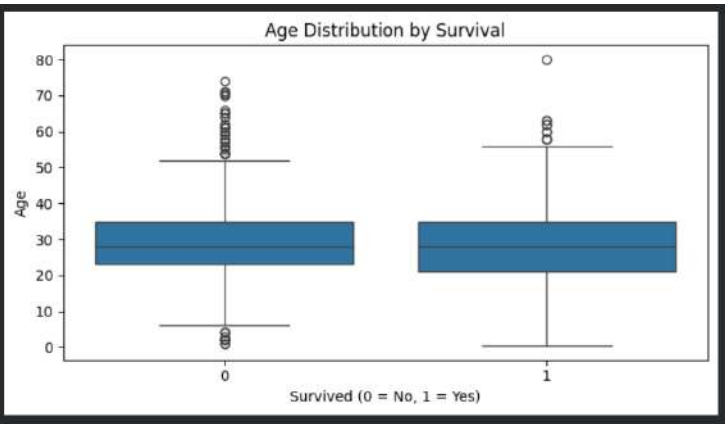
```
Survival rate by passenger class (%):
Pclass
1    62.96
2    47.28
3    24.24
Name: Survived, dtype: float64
```

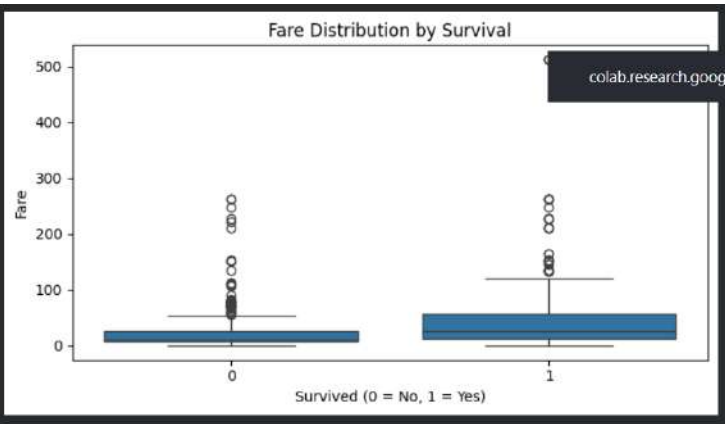
colab.research.google.com – To exit full screen, press Esc



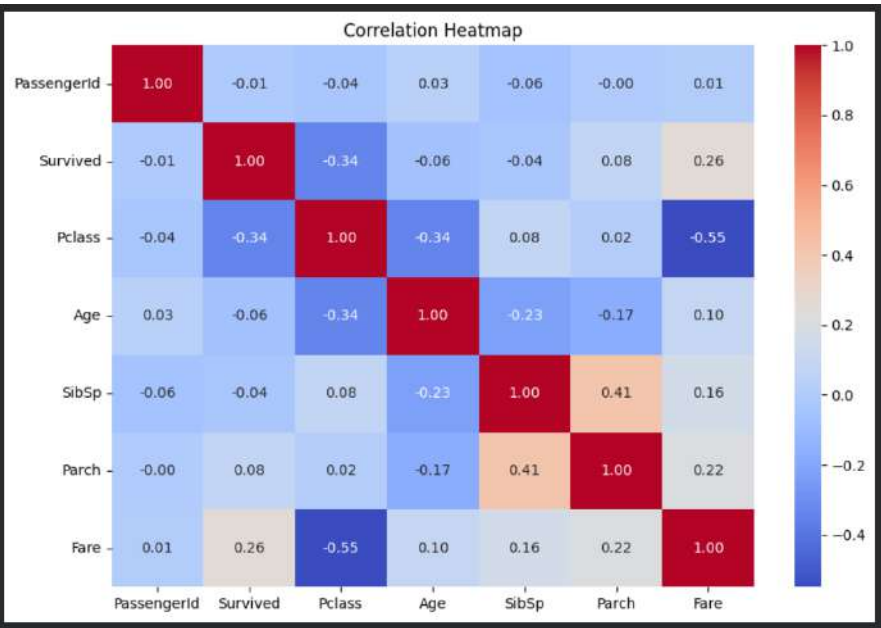


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Age Distribution by Survival
Age
80
70
60
50
40
30
20
10
0
0
1
Survived (0 = No, 1 = Yes)



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Correlation Heatmap
PassengerId
Survived
Pclass
Age
SibSp
Parch
Fare
1.00 -0.01 -0.04 0.03 -0.06 -0.00 0.01
-0.01 1.00 -0.34 -0.06 -0.04 0.08 0.26
-0.04 -0.34 1.00 -0.34 0.08 0.02 -0.55
0.03 -0.06 -0.34 1.00 -0.23 -0.17 0.10
-0.06 -0.04 0.08 -0.23 1.00 0.41 0.16
-0.00 0.08 0.02 -0.17 0.41 1.00 0.22
0.01 0.26 -0.55 0.10 0.16 0.22 1.00
1.0
0.8
0.6
0.4
0.2
0.0
−0.2
−0.4

```
Numerical features: ['Pclass', 'Age', 'SibSp', 'Parch', 'Fare']
Categorical features: ['Sex', 'Embarked']

Training records: 712
Testing records: 179

Training survival distribution:
Survived
0    0.617
1    0.383
Name: proportion, dtype: float64

Testing survival distribution:
Survived
0    0.615
1    0.385
Name: proportion, dtype: float64
```

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	Metric	Score
0	Accuracy	0.804469
1	Precision	0.793103
2	Recall	0.666667
3	F1 Score	0.724409

```
Project completed successfully.
```